



On superlevel sets of conditional densities and multivariate quantile regression

Annika Camehl^{*}, Dennis Fok, Kathrin Gruber

Econometric Institute, Erasmus University Rotterdam, Burgemeester Oudlaan 50 3062 PA Rotterdam, The Netherlands

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ABSTRACT

Some common proposals of multivariate quantiles do not sufficiently control the probability content, while others do not always accurately reflect the concentration of probability mass. We suggest superlevel sets of conditional multivariate densities as an alternative to current multivariate quantile definitions. Hence, the superlevel set is a function of conditioning variables much like in quantile regression. We show that conditional superlevel sets have favorable mathematical and intuitive features, and support a clear probabilistic interpretation. We derive the superlevel sets for a conditional or marginal density of interest from an (overfitted) multivariate Gaussian mixture model. This approach guarantees logically consistent (i.e., non-crossing) conditional superlevel sets and also allows us to obtain more traditional univariate quantiles. We demonstrate recovery of the true conditional univariate quantiles for distributions with correlation, heteroskedasticity, or asymmetry and apply our method in univariate and multivariate settings to a study on household expenditures.

1. Introduction

Regression quantiles (Koenker and Bassett, 1978) allow researchers to investigate relationships between variables not only at the center, but over the entire conditional distribution of a response. This advantage over conventional mean regression led to a rapidly expanding literature on quantile regression with countless applications in virtually all scientific disciplines (see, e.g., Koenker, 2017, for a comprehensive survey).

Now, an observation rarely comes as a single quantity and statistical analyses often require the exploration of joint quantiles of multiple response variables. The simplest approach is to decompose the multiple-output problem into individual sub-problems, each of which considers a different focal response variable as a function of the other variables (e.g., Ali et al., 2016). Although the information about the joint conditional dependence between the variables is retained, these quantiles are inherently one-dimensional and do not fully reflect the multivariate structure of the response. Vector-valued extensions to multiple-output regression quantiles aim to order the observed points in the multivariate data space, typically in relation to a global center, with smaller demarcated regions containing more central data points (see, e.g., Chaudhuri, 1996; Chakraborty, 2003; Hallin et al., 2010; Kong and Mizera, 2012). These definitions therefore do not collapse to the usual (i.e., left-to-right oriented) univariate quantiles in a one-dimensional setting and may instead better be referred to as *reference regions* (Hlubinka et al., 2022). The literature offers numerous such multivariate quantile proposals and the exact approaches differ widely in their properties and ease of implementation (see Serfling, 2002, a more recent review can be found in Hallin and Šiman, 2018). However, the properties of the quantile regions often very much depend on the underlying empirical distribution and thus, not all are guaranteed to cover a certain preset probability (Hallin

^{*} Corresponding author.

E-mail addresses: camehl@ese.eur.nl (A. Camehl), dfok@ese.eur.nl (D. Fok), gruber@ese.eur.nl (K. Gruber).

et al., 2010; Kong and Mizera, 2012). While the coverage problem is solved within elliptical (Hlubinka and Šíman, 2013; Hallin and Šíman, 2016) and center-outward quantile definitions (Carlier et al., 2016; Chernozhukov et al., 2017; Hallin et al., 2021; del Barrio et al., 2024), the determined center may lie well afield of the bulk of the data, in case the response variables feature non-linear dependencies or multi-modalities. Hence, the constructed quantile region can cover a large part of the support of a distribution with little or no probability mass (Zuo and Serfling, 2000; Carlier et al., 2016; Hallin et al., 2021; del Barrio et al., 2024). This turns the resulting statistical objects to be difficult to interpret in practice.

In this paper, we propose an alternative and consider the superlevel set of a *conditional* multivariate density for particular variables of interest, that is, the set of points for which the conditional density function exceeds a specific threshold (for the unconditional case, see Hartigan, 1987; Polonik, 1995). We set this threshold such that the set corresponds to the desired α probability coverage. Therefore, a probabilistic interpretation is automatically supported. Since we directly build on the multivariate distribution of the variables, we implicitly opt for the mode as the location measure. In case of multiple modes, we obtain a collection of disconnected regions and thus, deviate from the strictly single-center quantile definition. This way, even for multi-modal and highly asymmetric shapes, the superlevel set adapts to the data.

We embed our proposal of a (conditional) multivariate superlevel set in a three step estimation procedure: (1) fitting a general multivariate distribution to the observed data, (2) deriving a conditional distribution of interest, and (3) applying the superlevel set definition to this estimated conditional distribution. Note that one may also opt to condition on an empty variable set to effectively consider the full multivariate distribution. This way, we can construct multivariate as well as univariate superlevel sets for variables, or a variable, conditional on certain values for a set of other variables. Since all results are retrieved from a single, global distribution, inconsistencies, such as decreasing level sets at increasing α -level values, are avoided; and this also allows us to execute semi-parametric univariate quantile regression, in the traditional sense, without the risk of quantile crossing.

Various parametric and non-parametric methods can be used to estimate the multivariate density (e.g., Aitkin and Wilson, 1980; Silverman, 1986; Scott, 1992; McLachlan and Peel, 2000). However, we have two requirements: the calculation of marginal and conditional distributions given the estimation result, and of the (full, marginal and conditional) cumulative distribution function must be supported. We approach the density estimation task from a semi-parametric Bayesian perspective with a flexible prior for the unknown distribution (Escobar and West, 1995; Müller et al., 1996). Specifically, we opt for an overfitted finite mixture model with a hierarchical Dirichlet prior on the mixing weights (Nobile and Fearnside, 2007; Rousseau and Mengersen, 2011). This approach neither assumes a particular distribution nor does it require a (pre-set) bandwidth parameter choice. An efficient Markov Chain Monte Carlo algorithm makes it easy to sample the unknown parameters together with the number of components from a joint posterior (Malsiner-Walli et al., 2016), and in the end to quantify the parameter uncertainty in the estimates. Thus, we can retrieve multivariate and univariate superlevel sets, as well as conditional and marginal univariate quantiles in a coherent, computational attractive framework and provide an alternative to Bayesian considerations of multiple-output quantile regression (see Cai, 2010; Taddy and Kottas, 2010; Bhattacharya and Ghosal, 2021; Guggisberg, 2023).

The rest of the paper is organized as follows. In Section 2, we discuss our proposal of the (conditional) multivariate superlevel set in relation to prototypical one- and multidimensional quantile definitions, and establish its theoretical properties with regard to equivariance, nestedness, and uniqueness. In Section 3, we detail our setup of an overfitted finite mixture model for the joint distribution of the data. The construction principle of the superlevel set is discussed in Section 4. In Section 5, we evaluate the finite-sample performance of our overfitted mixture approach in relation to some selected benchmarks in a simulation study. Here, we focus on the univariate conditional quantile concept, due to the lack of competing estimation methods for the same multidimensional estimand. Section 6 illustrates our approach in the univariate and multivariate setting on a typical application from the economic literature. We conclude with a discussion in Section 7. Details on the sampling algorithm and additional simulation results are collected in the appendices. The computer codes are written in MATLAB (The Math Works Inc., 2020) and Julia (Bezanson et al., 2017), and are available at <https://github.com/aschnuecker/Superlevel-sets>.

2. Multivariate quantiles and superlevel sets

We begin with a discussion on different notions of multivariate quantiles. Thereby, we start with one-dimensional (i.e., scalar-valued) considerations and build up towards the multidimensional (i.e., vector-valued) case. Finally, we introduce our multivariate superlevel set proposal and establish its theoretical properties.

2.1. Scalar-valued approaches

The simplest notion of a multivariate quantile is that of a vector with the classical univariate quantiles as its components. Let $\mathbf{Y} = (Y_1, \dots, Y_K)' \in \mathbb{R}^K$ denote a K -dimensional random vector of response variables. The quantile of the k th random variable Y_k is the smallest value $Q_{Y_k}(\alpha)$ such that $\Pr[Y_k \leq Q_{Y_k}(\alpha)]$ is at least α :

$$Q_{Y_k}(\alpha) = \inf \left\{ y \in \mathbb{R} : F_{Y_k}(y) \geq \alpha \right\}, \quad \alpha \in (0, 1), \quad (1)$$

where $F_{Y_k}(\cdot)$ is the (right-continuously defined) cumulative distribution function of Y_k .

A dependency-bearing extension can be approached locally with the quantile of the k th response variable defined conditional on the remaining ones. Accordingly, let $\mathbf{Y}_{(-k)}$ denote the vector without the k th response variable. Thus, the conditional quantile

of Y_k is a function of the α -level and a (specified) value $y_{(-k)}$ of $Y_{(-k)}$:

$$\begin{aligned} Q_{Y_k|Y_{(-k)}=y_{(-k)}}(\alpha) &= \inf \left\{ y \in \mathbb{R} : F_{Y_k|Y_{(-k)}=y_{(-k)}}(y) \geq \alpha \right\} \\ &= \inf \left\{ y \in \mathbb{R} : \Pr[Y_k \leq y | Y_{(-k)} = y_{(-k)}] \geq \alpha \right\} \\ &= \inf \left\{ y \in \mathbb{R} : E[\rho_\alpha(Y_k - y) | Y_{(-k)} = y_{(-k)}] \right\}, \end{aligned} \quad (2)$$

where $F_{Y_k|Y_{(-k)}=y_{(-k)}}(\cdot)$ is the cumulative distribution function of Y_k conditional on the event that variable $Y_{(-k)}$ takes on value $y_{(-k)}$, and $\rho_\alpha(z) = (1 - \alpha)zI_{[z < 0]} + \alpha zI_{[z \geq 0]}$ is the standard check function (see Koenker and Bassett, 1978).

In the standard (i.e., Koenker–Bassett) quantile regression set-up, the conditional quantile $Q_{Y_k|Y_{(-k)}=y_{(-k)}}(\alpha)$ is parameterized as a linear function of $y_{(-k)}$, that is, $a + b'y_{(-k)}$. This is a global approach, and a and b are obtained by solving

$$\inf \{ E[\rho_\alpha(Y_k - a - b'Y_{(-k)})] : a \in \mathbb{R}, b \in \mathbb{R}^{K-1} \}, \quad (3)$$

with the expectation taken over Y_k and $Y_{(-k)}$. If Y follows a multivariate Gaussian, the solution to Eq. (3) coincides with that of Eq. (2), for every value of $y_{(-k)}$. However, if the distribution is not multivariate Gaussian, the same probability statement will not hold: in Eq. (3) the expectation is taken over Y , whereas in Eq. (2) the expectation is only taken over Y_k . Though, in the standard setting all “regressors” in the input-space are assumed fixed, in the context of multiple-output problems this assumption is not valid and instead, requires a vector-valued take on regression quantiles.

2.2. Vector-valued approaches

The directional adaptation of the univariate Koenker–Bassett approach defines the multivariate quantile as the intersection of a collection of sets (Hallin et al., 2010; Kong and Mizera, 2012). Included in this collection are not only the half-spaces for each of the K -variables, but also the half-spaces for all of their linear combinations. Consider the $K - 1$ dimensional unit sphere S^{K-1} , that is, the set of vectors in $\mathbb{R}^K$ with unit length. For direction $u \in S^{K-1}$, define the $K \times (K - 1)$ matrix of unit vectors Γ_u such that (u, Γ_u) is an orthonormal basis of $\mathbb{R}^K$. The α -level regression quantile of the univariate random variable $u'Y$ is obtained, in the (standard) Koenker–Bassett sense, from its orthogonal $K - 1$ dimensional random vector $\Gamma_u'Y$. Thus, we solve

$$\arg \inf \{ E[\rho_\alpha(u'Y - a - b'\Gamma_u'Y)] : a \in \mathbb{R}, b \in \mathbb{R}^{K-1} \}, \quad (4)$$

to generate a hyperplane $u'y = a_u + b_u'\Gamma_u'y$ for every $u \in S^{K-1}$ that splits $\mathbb{R}^K$ in two half-spaces, with α of the total probability mass on one side and $(1 - \alpha)$ probability mass on the other side. The hyperplane for the combination (α, u) equals the one for $(1 - \alpha, -u)$. The final multivariate quantile is the intersection of all half-spaces for $u \in S^{K-1}$. The separated region is bounded by the so-called directional quantile contour.

A more direct solution is to search for a unique ellipsoid with α -probability mass and to solve

$$\inf \{ E[\rho_\alpha(Y'AY + Y'b - c)] : A \in \mathbb{R}^{K \times K}, b \in \mathbb{R}^K, c \in \mathbb{R} \}, \quad (5)$$

subject to A being symmetric, positive semidefinite, and regularization constraint $\det(A) = 1$ (Hlubinka and Šiman, 2013; Hallin and Šiman, 2016). Here, the argument of ρ_α is negative if Y takes a value inside the ellipsoid, and positive otherwise. The optimal c , denoted c^* , is the α -level quantile of $Y'AY + Y'b$ such that for $c^* = c^*(A, b)$ it holds $\Pr[Y'AY + Y'b - c^* \leq 0] = \alpha$. Intuitively, the shape of this elliptical quantile is set to minimize the spread of the quadratic form $Y'AY + Y'b$ above and below the optimum c^* .

Alternative concepts consider an optimal mapping, from a reference distribution to a focal (conditional) distribution, to order the real-space from the center outward (Carlier et al., 2016; Chernozhukov et al., 2017; Hallin et al., 2021, an extension to conditional multivariate quantiles is presented in del Barrio et al., 2024). The reference distribution is chosen such that quantiles with correct probability coverage are straightforward to obtain. More formally, and largely following the notation of del Barrio et al. (2024), consider a focal K -dimensional random variable with (conditional) distribution P , and a K -dimensional reference random variable U with distribution U_K . The related center-outward quantile map satisfies $Q_\pm(U) \sim P$. This map is the unique gradient of a convex function from the K -dimensional open unit sphere S^K to $\mathbb{R}^K$. The reference distribution U_K is chosen such that its α quantile is given by the K -dimensional closed unit sphere αS^K with radius α . The α -level quantile of P is then $Q_\pm(\alpha S^K)$. In practice, the center-outward quantiles are obtained as the solution to a potentially large-scale linear programming problem.

Fig. 1 compares the directional Koenker–Bassett quantiles (areas within the gray lines indicating 20 different directions), the elliptical quantiles (areas within the black lines), and the center-outward quantiles (dashed red lines) for $\alpha = .8$ on two different bivariate Gaussian mixture distributions. The directional Koenker–Bassett quantiles have α -probability mass in each half-space, and the probability within the enclosed region will therefore be much smaller than $\alpha = .8$. For comparison, the elliptical quantile regions are constructed to exactly cover α and are much larger. The same holds for the center-outward quantiles. However, all three quantile definitions cover large parts of $\mathbb{R}^2$ with very low probability mass, which turns the multivariate statistical objects difficult to interpret in practice.

Alternatively, Wei (2008) proposes a multivariate quantile by considering the median of the distribution as a (global) center. The boundary of the corresponding quantile region is given by the α -coverage interval in a particular direction from the median and is parameterized with a smooth spline as a function of the angle of the direction. These quantiles are referred to as reference quantiles (an extension is provided in Hlubinka et al., 2022). Though, the resulting quantile covers α -probability mass, the dependence on the median limits its applicability to unimodal distributions. Likewise, for the examples in Fig. 1, this definition of a directional quantile will yield regions containing large areas with low probability mass, since the density at the multivariate median is close to zero.

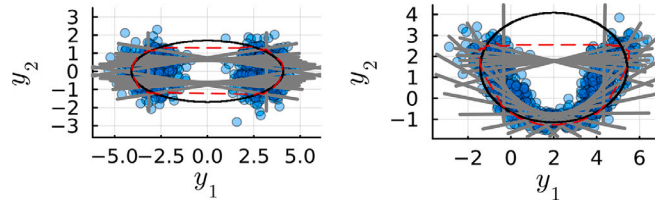


Fig. 1. Directional Koenker–Bassett, elliptical and center-outward quantile contours for two examples of non-Gaussian data. Scatter plots include 1000 simulated observations. Areas enclosed within the gray lines (—) give the 80% directional quantile (20 different directions); within the black lines (—) the 80% elliptical quantile, and within the dashed red lines (---) the 80% center-outward quantile.

2.3. The superlevel-set approach

We aim to create an alternative to the earlier mentioned multivariate quantiles that reflects the concentration of probability mass and has guarantees on the covered probability. To formalize this, we search for the smallest possible region that covers the given probability content. This makes the corresponding multivariate object a property of the density function of a random variable.

First, we introduce some notation. Let $\mathcal{K} \subseteq \{1, 2, \dots, K\}$ be the set of indices of focal response variables, and $C \subset \{1, 2, \dots, K\}$ the indices of the variables collected in the conditioning set, where $\mathcal{K} \neq \emptyset$ and $\mathcal{K} \cap C = \emptyset$.¹ $|\mathcal{K}| > 1$ gives a multivariate distribution, and C might be equal to $\{1, \dots, K\} \setminus \mathcal{K}$, $\emptyset$, or something in between. $\mathcal{K}$ or C in a subscript selects the corresponding elements from a vector, matrix or function. Accordingly, the density of $Y_{\mathcal{K}}$ given $Y_C = y_C$ is $f_{Y_{\mathcal{K}}|Y_C=y_C}(y_{\mathcal{K}}|y_C)$. Then, the problem of finding the smallest possible region with probability coverage α is:

$$\begin{aligned} Q_{Y_{\mathcal{K}}|Y_C=y_C}(\alpha) &= \underset{Q}{\operatorname{argmin}} |Q|, \\ \text{subject to } \Pr[Y_{\mathcal{K}} \in Q | Y_C = y_C] &\geq \alpha. \end{aligned} \quad (6)$$

Here, Q is considered an open set. Note that with the conditional formulation we acknowledge a researcher's (usual) interest in the conditional multivariate distribution. For $C = \emptyset$, we obtain the unconditional region based on the unconditional multivariate distribution, and for $\mathcal{K} = \{1, \dots, K\}$, we obtain the joint region of the full random vector.

In Appendix A, we prove that for a continuous conditional density the solution to Eq. (6) is the smallest (strict) superlevel set of the conditional density with at least α probability: if a point outside the set would have a higher density than a point inside, one can remove an area around the inside point from the set and replace this by a smaller area around the outside point. Appropriate area choices keep the probability content the same but reduce the volume of the set which contradicts the minimum volume property.

We continue with the superlevel-set definition (hereinafter also referred to as SLS) and name the resulting object as such. The SLS is also properly defined for discrete random variables and even random variables that are discrete-continuous mixtures. The smallest-possible-region formulation then requires a definition of the volume of discrete points.

Let $\mathcal{Y}_{\mathcal{K}}(t|y_C)$ be the set of values $y_{\mathcal{K}} \in \mathbb{R}^{|\mathcal{K}|}$ such that the conditional distribution of $Y_{\mathcal{K}}$ given $Y_C = y_C$ exceeds t :

$$\mathcal{Y}_{\mathcal{K}}(t|y_C) = \{y_{\mathcal{K}} : f_{Y_{\mathcal{K}}|Y_C=y_C}(y_{\mathcal{K}}|y_C) > t\}. \quad (7)$$

The SLS covers α -probability mass with the smallest possible set:

$$\begin{aligned} Q_{Y_{\mathcal{K}}|Y_C=y_C}(\alpha) &= \mathcal{Y}_{\mathcal{K}}(t_{\alpha}^*|y_C), \\ \text{where } t_{\alpha}^* &= \sup\{t : \Pr[Y_{\mathcal{K}} \in \mathcal{Y}_{\mathcal{K}}(t|y_C) | Y_C = y_C] \geq \alpha\}. \end{aligned} \quad (8)$$

The collection of superlevel sets for all values of α characterizes the entire underlying distribution, that is, it is possible to retrieve the density level from the probability content of the SLS collection. For continuous densities, the density level of points in the difference of two superlevel sets with probability content p and $p + \epsilon$ is constant for $\epsilon > 0$ small enough. The respective density level is then ϵ divided by the volume of the difference.

Whether a probability content equal to α is exactly attainable, depends on the smoothness of $\Pr[Y_{\mathcal{K}} \in \mathcal{Y}_{\mathcal{K}}(t|y_C) | Y_C = y_C]$ as a function of t . This function will be non-smooth in case the conditional density of $Y_{\mathcal{K}}$ contains areas with constant density: a small increase in t then leads to a jump in $\Pr[Y_{\mathcal{K}} \in \mathcal{Y}_{\mathcal{K}}(t|y_C) | Y_C = y_C]$. In case the jump is located such that $\Pr[Y_{\mathcal{K}} \in \mathcal{Y}_{\mathcal{K}}(t|y_C) | Y_C = y_C] = \alpha$ has no solution, $Q_{Y_{\mathcal{K}}|Y_C=y_C}(\alpha)$ will contain more than α -probability mass. In the case of the conditional density being uniform on $[0, 1]^{|\mathcal{K}|}$, we have $Q_{Y_{\mathcal{K}}|Y_C=y_C}(\alpha) = [0, 1]^{|\mathcal{K}|}$ for all $\alpha \in [0, 1]$. The latter extreme situation will not occur often in practice and most of the time we have $\Pr[Y_{\mathcal{K}} \in Q_{Y_{\mathcal{K}}|Y_C=y_C}(\alpha) | Y_C = y_C] = \alpha$. Areas with constant density lead to multiple solutions to the minimization problem in Eq. (6). Then, the definition of the minimum set is still equivalent to that of the superlevel set, since the argmin operator gives the union of all solutions. The exact probability coverage of $Q_{Y_{\mathcal{K}}|Y_C=y_C}(\alpha)$ is easy to obtain. In fact, it is a by-product of the algorithm in Section 4.1.

¹ We do not require $\mathcal{K} \cup C = \{1, 2, \dots, K\}$.

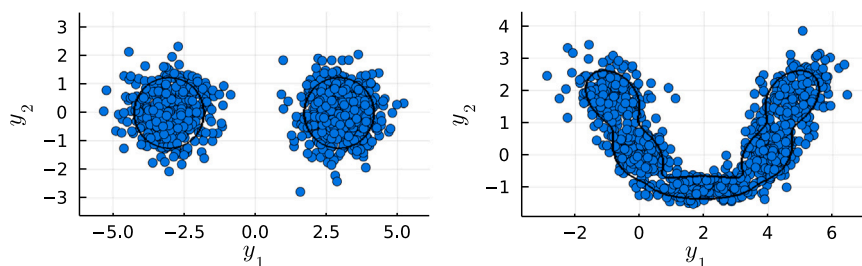


Fig. 2. Superlevel set contours for two examples of non-Gaussian data. Scatter plots include 1000 simulated observations from two bivariate distributions. Areas enclosed within the black lines (—) correspond to 80% probability mass.

Fig. 2 shows the 80% SLS contours for the two bivariate mixture distributions presented in Section 2.2. In both scenarios, the bimodal and the asymmetric distribution, we obtain flexibly shaped α -dependent contours. For a multimodal distribution, $Q_{Y_K|Y_C=y_C}(\alpha)$ is generally a union of open sets; and for a distribution which features nonlinear dependencies, the shape of $Q_{Y_K|Y_C=y_C}(\alpha)$ can also be far from elliptical.

2.4. Properties

The proposed multivariate superlevel set satisfies practically relevant properties: it ensures a clear coverage probability interpretation, reflects the concentration of probability mass, and the shape of the SLS naturally adapts to various distributional forms. Moreover, it enjoys many desirable theoretical properties of a well-behaved quantile concept, that is, nestedness, uniqueness, and affine equivariance (see Serfling, 2002; Hallin and Šiman, 2018). We prove these properties at the end of the section.

Table 1 summarizes the properties of the (conditional) multivariate superlevel set in comparison to the discussed multivariate quantile proposals: (1) a collection of classic univariate quantiles with each focal response variable considered as a function of the other variables in the output-vector; (2) the directional Koenker–Bassett quantile (Hallin et al., 2010; Kong and Mizera, 2012); (3) the elliptical quantile (Hlubinka and Šiman, 2013; Hallin and Šiman, 2016); (4) the center-outward quantile (Hallin et al., 2021; del Barrio et al., 2024); and (5) the reference quantile as proposed by Wei (2008). The table shows that the superlevel-set concept is more than competitive.

The collection of univariate quantiles ignores the multi-dimensionality of the statistical problem. Consequently, almost all of the practically desirable properties do not hold. These scalar-valued quantiles are not even guaranteed to be nested and the quantile contours can cross at different α -levels. The nestedness property is ensured within all vector-valued extensions. However, the directional Koenker–Bassett quantile yields statistical objects that are difficult to interpret: the probability coverage is not known beforehand and the enclosed areas may not accurately reflect the concentration of probability mass in the data. The coverage

Table 1
Properties of the multivariate superlevel set in comparison to different multivariate quantile proposals.

	Univariate quantile collection	Directional quantile	Elliptical quantile	Center- outward quantile	Reference quantile	Superlevel set
Vector-valued	No	✓	✓	✓	✓	✓
α -probability coverage control	No	No	✓	✓	✓	✓
Reflects probability mass concentration	No	No	No	No	No	✓
Allows for disconnected sets	No	No	No	No	No	✓
Nestedness	No	✓	✓	✓	✓	✓
Uniqueness	✓	✓	✓	✓	✓	✓
Affine equivariance	No	✓	✓	No	✓	✓
Invariant to coordinate- wise transformations	✓	No	No	No	No	No
Invariant to monotone transformations	No	No	No	No	No	No
Reduces to univariate left-to-right quantile	✓	No	No	No	no	No
Reduces to univariate interquantile interval	✓	No	No	✓	✓	No

problem is resolved with the elliptical, the center-outward and the reference quantile concepts. Still, the quantile regions remain misleading in terms of the location of the probability mass. The multivariate superlevel set is the only approach with guarantees on the covered probability, and an adequate reflection of probability mass.

Generally, all vector-valued quantiles – except the center-outward quantiles due to relying on the Euclidean distance – are invariant to affine but not to coordinate-wise or general monotone transformations. The directional Koenker–Bassett, the elliptical, and center-outward quantiles are convex by design and a convexity violating monotone transformation is easy to construct. Likewise, similar counter-examples exist for the reference quantile. We agree with Carlier et al. (2016) that it is not possible to reproduce all properties of scalar quantile regression in higher dimensions. The superlevel set is very much like the *highest posterior density* (HPD) interval, commonly used in Bayesian statistics to quantify estimation uncertainty in model parameters (Box and Tiao, 1965; Wei, 2008). For a univariate, unimodal posterior, the HPD interval coincides with the SLS of the posterior density. For univariate, multimodal posteriors, however, the HPD is still operationalized as an interval, while the SLS gives a set of intervals. The exact parameterization of a model is, to some extent, arbitrary and thus, transformation invariance is desired for posterior intervals for parameters (see Bernardo, 2005). Contrary, response variables are measured in specific real-world units, for which invariance is less useful in practice. We therefore do not see the lack of invariance as a true problem. In fact, it is a natural consequence of our desire to reflect the concentration of probability mass in the data.

Finally, none of the vector-valued quantiles collapses to the traditional (left-to-right oriented) quantile in one dimension, while some reduce to the interquantile interval, that is, $[Q(\frac{1-\alpha}{2}), Q(\frac{1+\alpha}{2})]$ with probability content α . The SLS does not rely on a global center in order to allow for disconnected sets. However, this implies that the SLS is not guaranteed to be cumulative and neither reduces to the traditional quantile nor to the interquantile interval in the univariate case. This is the trade-off between adequate reflection of the probability mass and connected sets.

Next, we prove that the SLS possess the properties of nestedness, uniqueness, and affine equivariance.

Property 1 (nestedness). *The SLS collection forms a sequence of nested sets such that $Q_{Y_K|Y_C=y_C}(\alpha_1) \subseteq Q_{Y_K|Y_C=y_C}(\alpha_2)$ for $0 < \alpha_1 \leq \alpha_2 < 1$ and all $y_C \in \mathbb{R}^{|C|}$.*

Proof. The nestedness property follows directly from the construction of the SLS. The set $\mathcal{Y}_K(t|y_C)$ increases with decreasing t , and therefore $\Pr[Y_K \in \mathcal{Y}_K(t|y_C)|Y_C = y_C]$ does not decrease as t decreases. Thus, it follows $t_{\alpha_2}^* \leq t_{\alpha_1}^*$ and we obtain $Q_{Y_K|Y_C=y_C}(\alpha_1) = \mathcal{Y}_K(t_{\alpha_1}^*|y_C) \subseteq \mathcal{Y}_K(t_{\alpha_2}^*|y_C) = Q_{Y_K|Y_C=y_C}(\alpha_2)$. $\square$

Property 2 (uniqueness). *Only one subset of $\mathbb{R}^{|K|}$ satisfies the definition of the multivariate object $Q_{Y_K|Y_C=y_C}(\alpha)$ for a given α and y_C .*

Proof. Assume there are two distinct sets $\mathcal{Y}_1$ and $\mathcal{Y}_2$ that satisfy Eq. (8). If both corresponding threshold values are identical, the sets are identical. If the threshold values are not identical, it holds $t_{\alpha_1}^* > t_{\alpha_2}^*$ or $t_{\alpha_1}^* < t_{\alpha_2}^*$, and the sets $\mathcal{Y}_1$ and $\mathcal{Y}_2$ cannot both satisfy the definition of the multivariate SLS. $\square$

Property 3 (affine equivariance). *For a vector a and a given invertible matrix A , both of appropriate size: $y_K \in Q_{Y_K|Y_C=y_C}(\alpha) \iff a + Ay_K \in Q_{a+AY_K|Y_C=y_C}(\alpha)$ for all $y_K \in \mathbb{R}^{|K|}$, $y_C \in \mathbb{R}^{|C|}$ and $\alpha \in (0, 1)$.*

Proof. Let $f_{Y_K|Y_C=y_C}(\cdot)$ denote the conditional density of Y_K , and $f_{a+AY_K|Y_C=y_C}(\cdot)$ denote the conditional density of the transformed random vector $a + AY_K$. Next, let t_{α}^* be the α -level threshold of Y_K and define $\tilde{t} = |A|^{-1}t_{\alpha}^*$. Given that $f_{a+AY_K|Y_C=y_C}(a + Ay_K) = |A|^{-1}f_{Y_K|Y_C=y_C}(y_K)$ for any $y_K \in \mathbb{R}^{|K|}$ and $y_C \in \mathbb{R}^{|C|}$:

$$\begin{aligned} y_K \in \mathcal{Y}_{Y_K}(t_{\alpha}^*|y_C) &\iff f_{Y_K|Y_C=y_C}(y_K) > t_{\alpha}^*, \\ &\iff f_{a+AY_K|Y_C=y_C}(a + Ay_K)|A| > t_{\alpha}^* = |A|\tilde{t}, \\ &\iff f_{a+AY_K|Y_C=y_C}(a + Ay_K) > \tilde{t}, \\ &\iff a + Ay_K \in \mathcal{Y}_{a+AY_K}(\tilde{t}|y_C). \end{aligned}$$

It follows $\Pr[a + AY_K \in \mathcal{Y}_{a+AY_K}(\tilde{t}|y_C)|Y_C = y_C] = \Pr[Y_K \in \mathcal{Y}_{Y_K}(t_{\alpha}^*|y_C)|Y_C = y_C]$ such that $Q_{a+AY_K|Y_C=y_C}(\alpha) = a + AQ_{Y_K|Y_C=y_C}(\alpha)$. Consequently, the SLS is equivariant under invertible linear transformations. $\square$

3. Finite mixture regression

We proceed with the estimation of the multivariate density for observed data. Therefore, we aim to describe the joint distribution of K response variables given a common set of G additional covariates for N observations with a flexible multivariate model. We also allow for the absence of covariates, that is, $G \geq 0$. Henceforth, let $y_n = (y_{n1}, \dots, y_{nK})$ be the K -dimensional output-vector of response variables and $x_n = (x_{n1}, \dots, x_{nG})$ be the G -dimensional input-vector of covariates for the n th observation unit. We consider the distribution of y_n to be an M -component mixture of multivariate Gaussian densities for which the component means can be a function of x_n :

$$f(y_n|x_n) = \sum_{m=1}^M \kappa_m \phi(y_n; g_m(x_n), \Sigma_m), \quad (9)$$

with scalar-valued component mixing weights $\{\kappa_m\}$, subject to constraints $\kappa_m \geq 0$ and $\sum_{m=1}^M \kappa_m = 1$, component-specific regression functions:

$$\mathbf{g}_m(\mathbf{x}_n) = \boldsymbol{\mu}_m + \mathbf{B}_m \mathbf{x}_n, \quad (10)$$

with location vectors $\{\boldsymbol{\mu}_m\}$ of dimension $K \times 1$, coefficient matrices $\{\mathbf{B}_m\}$ of dimension $K \times G$, component-specific error covariance matrices $\{\boldsymbol{\Sigma}_m\}$ of dimension $K \times K$; and density function $\phi(\cdot)$.

Note that our interest is in the properties of the response variables in $\mathbf{y}_n$, and not in those of the exogenous variables in $\mathbf{x}_n$. We therefore differentiate between $\mathbf{y}_n$ and $\mathbf{x}_n$ in the conditioning set: For a subset of K response variables, we define the distribution for $\mathbf{y}_K$ conditional on $\mathbf{x}$ and $\mathbf{y}_C$. Hence, $\mathbf{y}_n$ depends linearly on $\mathbf{x}_n$. This reduces the complexity of the problem, since the dimension of the multivariate density only depends on $\mathbf{y}_n$.

The Gaussian mixture formulation is quite general and, given the information encoded in the exogenous variables, covers a wide range of multivariate models. For $M > 1$, we obtain a flexible parameterization and for $M = 1$, a multivariate linear formulation is recovered. The number of components is treated as unknown and estimated jointly with the component-specific parameters from the data. Note that other component distribution choices, e.g., a multivariate skew-Gaussian or a multivariate skew-t (e.g., Lin et al., 2007; Frühwirth-Schnatter and Pyne, 2010), are possible as well. However, given enough Gaussian components, large classes of conditional multivariate densities can be approximated, with sufficient accuracy (Roeder and Wasserman, 1997). Though the assumption of a Gaussian base-distribution restricts the specified model to continuous response variables, this limitation does not apply to the definition of the multivariate superlevel set.

3.1. Prior distributions

We assume an intentionally overfitted model with a finite but comparatively large number of components. The mixture is reduced to the final model through a sparse hierarchical prior on the collection of possible density functions. Accordingly, we specify a hierarchical Dirichlet prior for the component mixing weights:

$$\boldsymbol{\kappa} | \{\bar{\rho}_m\} \sim D(\bar{\rho}_1, \dots, \bar{\rho}_M), \quad (11)$$

where the component-specific concentration parameters $\{\bar{\rho}_m\}$ follow a Gamma distribution:

$$\bar{\rho}_m \sim \mathcal{G}(\underline{a}_1, 1/(\underline{a}_2 M)), \quad (12)$$

with hyperparameters $\underline{a}_1, \underline{a}_2 > 0$. The component locations are given a multivariate Gaussian prior:

$$\boldsymbol{\mu}_m | \bar{\mathbf{v}}_0, \bar{\mathbf{V}}_0 \sim \mathcal{N}(\bar{\mathbf{v}}_0, \bar{\mathbf{V}}_0), \quad (13)$$

with an improper Gaussian prior placed on the component prior mean $\bar{\mathbf{v}}_0$ and a hierarchical prior on the component prior variance $\bar{\mathbf{V}}_0$:

$$\bar{\mathbf{v}}_0 \sim \mathcal{N}(\underline{\mathbf{v}}, \underline{\mathbf{V}}), \quad (14)$$

$$\bar{\mathbf{V}}_0 = \text{diag}(R_1^2 \lambda_1, \dots, R_K^2 \lambda_K), \quad (15)$$

$$\lambda_k \sim \mathcal{G}(\underline{b}_1, 1/\underline{b}_2), \quad (16)$$

where $\underline{\mathbf{v}} = \text{median}(\mathbf{y}_n)$ and $\underline{\mathbf{V}}^{-1} = \mathbf{0}$. $\{R_k\}$ correspond to the response-variable-specific value ranges and $\{\lambda_k\}$ are local shrinkage factors following a Gamma distribution with hyperparameters $\underline{b}_1, \underline{b}_2 > 0$ (see Brown and Griffin, 2010). We complete our model with a multivariate Gaussian prior for the vector of component-specific regression coefficients: $\text{vec}(\mathbf{B}_m) \sim \mathcal{N}(\mathbf{c}_0, \mathbf{C}_0)$ with prior mean $\mathbf{c}_0$ and prior variance $\mathbf{C}_0$; and a conjugate inverse-Wishart prior for the component-specific error covariance matrices: $\boldsymbol{\Sigma}_m \sim \text{IW}(\mathbf{S}_0, s_0)$ with scale matrix $\mathbf{S}_0 = \mathbf{I}$ and degrees of freedom $s_0 > 2 + K$, to rule out variances equal or close to zero and to ensure a proper posterior.

Asymptotically, the hierarchical Dirichlet prior on the mixing weights (Eq. (11)–(12)) matches a Dirichlet process prior with expectation $E(\bar{\rho}_m) = 1/M$ and variance $\text{Var}(\bar{\rho}_m) = 1/(\underline{a}_2 M^2)$ (see Ishwaran et al., 2001). Thus, as the number of mixture distributions M increases, the relative size of the component mixing weights $\{\bar{\rho}_m\}$ decreases, and the sequence of Gaussian mixtures becomes increasingly sparse. The hyperparameters $\underline{a}_1$ and $\underline{a}_2$ have a regularizing effect such that the posterior distribution of the weights is encouraged to concentrate at zero for superfluous components. The multivariate Normal-Gamma prior (Eq. (14)–(16)) then puts strong shrinkage on the component means via the prior variance, particularly for heavily overlapping densities, to reduce the bias (caused by spurious components) for components with a small number of observations (see also Richardson and Green, 1997; Stephens, 2000; Yau and Holmes, 2011).

3.2. Posterior inference

Given data on N observations, we obtain posterior results via a data-augmented MCMC algorithm (Diebolt and Robert, 1994). Thus, we associate each observation $\mathbf{y}_n$ to a latent allocation variable $z_n \in \{1, \dots, M\}$ such that:

$$f(\mathbf{y}_n | z_n = m, \mathbf{x}_n) = \phi(\mathbf{y}_n; \mathbf{g}_m(\mathbf{x}_n), \boldsymbol{\Sigma}_m), \quad (17)$$

$$\Pr[z_n = m] = \kappa_m. \quad (18)$$

Exact details on the full conditionals and the sampling steps are provided in [Appendix B](#). After each MCMC iteration a random permutation step is added to ensure that the sampler explores all modes of the full posterior distribution ([Frühwirth-Schnatter, 2001](#)). Most parts of this algorithm are quite standard with two exceptions: the full conditional distribution for the shrinkage factors ([Malsiner-Walli et al., 2016](#)) and the multivariate random-walk Metropolis–Hastings step for the component-specific concentration parameters.

In case component-specific posterior means or variances are needed, the label-switching problem present in most mixture models has to be solved. We identify the component labels in a separate post-processing procedure. We cluster the component-specific posterior MCMC samples using a K-centroids Mahalanobis distance-based algorithm to reduce the dimensionality of the relabeling problem and to capture elliptical shapes and different volumes of the posteriors (a detailed description is given in [Malsiner-Walli et al., 2016](#)). This identification scheme can easily be applied to non-Gaussian mixture settings as well.

4. Implementation

Given samples from the posterior of the Gaussian mixture model, we can obtain posterior estimates for Y_K given $Y_C = y_C$, either at the posterior means or for each posterior draw. The latter allows a straightforward evaluation of the posterior uncertainty in the estimates, while the former involves lower computational costs. Note that the use of posterior means requires the previously discussed label-switching problem to be solved.

Specifically, we aim to construct multivariate and univariate superlevel sets and/or traditional univariate quantiles given some value of the other response variables, and some value for the covariates. Accordingly, the conditional distribution of Y_K given the estimated model parameters collected in Θ , concrete values for the variables in the output-vector y_C , and the covariates in the input-vector x , is needed. It is easy to show that this conditional distribution is again an M -component mixture of Gaussians:

$$f_{Y_K|Y_C=y_C}(y_K|y_C, x, \Theta) = \sum_{m=1}^M \omega_m^C(y_C, x) \phi(y_K; \mu_m^{K|C}(y_C, x), \Sigma_m^{K|C}), \quad (19)$$

with component-specific conditional mean $\mu_m^{K|C}(y_C, x)$ and component-specific conditional variance $\Sigma_m^{K|C}$:

$$\mu_m^{K|C}(y_C, x) = g_{m,K}(x) + \Sigma_{m,K,C}^{-1} \Sigma_{m,C,C} (y_C - g_{m,C}(x)), \quad (20)$$

$$\Sigma_m^{K|C} = \Sigma_{m,K,K} - \Sigma_{m,K,C} \Sigma_{m,C,C}^{-1} \Sigma_{m,C,K}, \quad (21)$$

and (updated) conditional mixing weight ω_m^C defined in terms of the Gaussian components in the conditioning set:

$$\omega_m^C(y_C, x) = \frac{\kappa_m \phi(y_C; g_{m,C}(x), \Sigma_{m,C,C})}{\sum_{l=1}^M \kappa_l \phi(y_C; g_{l,C}(x), \Sigma_{l,C,C})}. \quad (22)$$

The constructed conditional distribution serves as input for the respective computational procedures in the following subsections.

4.1. Multivariate superlevel sets

The conditional Gaussian mixture distribution at given parameter values is used to numerically solve the optimization problem that defines the multivariate superlevel set (see Eq. (8)). The procedure is summarized in Algorithm 1.

First, we select a subset of $\mathbb{R}^{|\mathcal{K}|}$ with a sufficiently large probability mass: We take the Cartesian product of the intervals $[F_{Y_k|Y_C=y_C}^{-1}(\epsilon), F_{Y_k|Y_C=y_C}^{-1}(1-\epsilon)]$ for each $k \in \mathcal{K}$, where ϵ is a small positive number and $F_{Y_k|Y_C=y_C}^{-1}(\cdot)$ is the quantile function for variable Y_k given $Y_C = y_C$ (see Eq. (19)). Then, we place n_{grid} grid points on each interval, running from the corresponding ϵ -marginal to the $(1-\epsilon)$ -marginal quantile, to split the subset into $(n_{\text{grid}} - 1)^{|\mathcal{K}|}$ hypercubes. Next, we calculate the enclosed probability content for each hypercube based on the estimated cumulative density functions. Finally, we add hypercubes in the order of the highest probability to $\tilde{Q}_{Y_K|Y_C=y_C}(\alpha)$ until the chosen α -coverage probability is exceeded or (exactly) reached. Hypercubes with the same probability are added simultaneously.

The discretization of the probability space results in a numerical approximation of the multivariate superlevel set. This approximation is accurate as long as the number of grid points is sufficiently large. Otherwise, the enclosed probability content of a hypercube may be too large and the resulting SLS boundaries will be non-smooth. To avoid hypercubes at the border of the grid, ϵ should be set to a small value. In [Fig. 2](#), for instance, we set $\epsilon = 0.001$ and $n_{\text{grid}} = 250$. Note, that a too small ϵ value and/or a too high grid point number will increase the necessary computation time.

4.2. Conditional univariate quantiles

Likewise, we can construct conditional univariate quantiles, in the traditional sense, directly from the (estimated) conditional distributions. For the conditional Gaussian mixture distribution, we obtain the univariate quantiles numerically as the solution to a simple root-finding problem. Algorithm 2 provides a stylized description of the procedure on the example of a variable k conditional on the values of all variables in C , where $k \notin C$.

First, we partition the output-vector and the corresponding mean and error variance estimates to calculate the conditional component-specific moments $\mu_m^{k|C}(y_C, x)$ and $\Sigma_m^{k|C}$ (Eq. (20)–(21)), and the conditional mixing weights $\omega_m^C(y_C, x)$ (Eq. (22)). Next, we derive the α -level quantile from the distribution function of the conditional Gaussian mixture, that is, we find the value q for which $F_{Y_k|Y_C=y_C}(q) = \alpha$ with a standard bisection method. The conditional component quantiles specify the interval for the root-finder, with the smallest and largest component quantiles providing the lower and upper bounds, respectively.

Algorithm 1: Calculation of multivariate superlevel-set

Input : chosen coverage probability α
conditional distribution function $F_{Y_K|Y_C=y_C}(\mathbf{y})$
grid boundary probability ϵ
dimension-specific grid point number n_{grid}

Output: actual coverage probability p
numerical SLS $\tilde{Q} = \tilde{Q}_{Y_K|Y_C=y_C}(\alpha)$ of size $n_{\text{grid}}^{|\mathcal{K}|}$

```

1 for  $k \in \mathcal{K}$  do
2    $\text{grid}_k =$  equally spaced  $n_{\text{grid}}$  vector with values
   from  $F_{Y_k|Y_C=y_C}^{-1}(\epsilon)$  to  $F_{Y_k|Y_C=y_C}^{-1}(1 - \epsilon)$ ;
3    $\tilde{Q}_{Y_K|Y_C=y_C}(\alpha) = |\mathcal{K}|$ -dimensional array of zeros
4    $P =$  empty  $|\mathcal{K}|$ -dimensional array to hold probabilities per hypercube
5   for  $(i_1 \in 2 : n_{\text{grid}}), (i_2 \in 2 : n_{\text{grid}}), \dots, (i_{|\mathcal{K}|} \in 2 : n_{\text{grid}})$  do
6      $P_{i_1, i_2, \dots, i_{|\mathcal{K}|}} = \Pr[Y_k \in [\text{grid}_{k, i_k-1}, \text{grid}_{k, i_k}] \forall k \in \mathcal{K} | Y_C = \mathbf{y}_C]$ 
7    $p = 0$ 
8   while  $p < \alpha$  do
9      $\mathcal{I} =$  set of indices for which  $P$  equals  $\max\{P\}$ 
10     $p = p + \sum_{i \in \mathcal{I}} P_i$ 
11    for  $i \in \mathcal{I}$  do
12       $\tilde{Q}_i = \alpha, P_i = 0$ 
```

Algorithm 2: Calculation of conditional univariate quantiles

Input : chosen coverage probability α
index of focal response variable k
value of response variables $\mathbf{y}_C$
estimated component-specific parameters $\{g_m(\mathbf{x}_n)\}, \{\Sigma_m\}, \{\kappa_m\}$

Output : conditional univariate quantile $Q_{k|Y_C=y_C}(\alpha)$

```

1 for  $m \in \{1, \dots, M\}$  do
2   Calculate conditional moments  $\bar{x}_m = \mu_m^{k|C}(\mathbf{y}_C, \mathbf{x})$  and  $s_m^2 = \Sigma_m^{k|C}$ 
3   Calculate component-specific quantiles  $q_m = \mu_m^{k|C}(\mathbf{y}_C, \mathbf{x}) + \Sigma_m^{k|C} \Phi^{-1}(\alpha)$ 
4   Calculate conditional weight  $w_m = \phi_m^C(\mathbf{y}_C, \mathbf{x})$ 
5 for all  $k \in \mathcal{K}$  do
6   Define  $h(q) = \sum_{m=1}^M w_m \Phi((q - \bar{x}_m) / \sqrt{s_m^2}) - \alpha$ 
7   Use bisection to find root of  $h(q)$  starting with interval  $[\min_m q_m, \max_m q_m]$ 
```

$\Phi(\alpha)$ and $\Phi^{-1}(\alpha)$ are the standard Gaussian distribution function and its inverse.

4.3. Quantile-specific measures

Since inference for any set of α -level values is based on a global posterior, contour crossing (i.e., the problem of non-monotonicity of the SLS or quantile contours) is automatically avoided. Hence, we can infer valid superlevel sets or quantiles for any (linear) combination of response variables and likewise, create a simple model of treatment and control for marginal SLS or quantile effects.

We define the local marginal effect in the α -level univariate quantile of response variable Y_k conditional on $Y_C = \mathbf{y}_C$ for a change from $\mathbf{x}$ to $\mathbf{x} + \Delta_g$:

$$\beta_{k|C}^g(\alpha | \mathbf{y}_C, \mathbf{x}) = Q_{Y_k|Y_C=y_C}(\alpha | \mathbf{x} + \Delta_g) - Q_{Y_k|Y_C=y_C}(\alpha | \mathbf{x}), \quad (23)$$

where Δ_g is a vector with a small value δ_g at position g and zeros elsewhere (Doksum, 1974). Alternatively, the marginal effect can be considered a derivative with δ_g dividing the result of Eq. (23). Similar marginal effects can be defined for changes in the conditioning variables $\mathbf{y}_C$ in a straightforward way.

The derived univariate marginal quantile effects permit a direct local interpretation at a certain value of $\mathbf{x}$ and $\mathbf{y}_C$, or a global interpretation through a (weighted) average across all observations, $\frac{1}{N} \sum_{n=1}^N \beta_{k|C}^g(\alpha | \mathbf{y}_{n,C}, \mathbf{x}_{n,C})$. The computation of the corresponding interval estimates for posterior uncertainty quantification is straightforward: Eq. (23) is evaluated for each posterior draw and the results are summarized either in terms of a posterior variance or the highest posterior density region.

5. Simulation study

We compare the performance of our Gaussian mixture-based univariate quantile estimates to established approaches in a Monte Carlo simulation study. Note that the lack of a generally accepted multivariate quantile definition restricts this simulation exercise to univariate conditional considerations. We cannot label the superlevel set as the objective truth and ensure a fair comparison with other multivariate quantile concepts. Still, a good fit of univariate quantiles based on conditional Gaussian mixture distributions readily translates into good estimation performance for the multivariate superlevel set.

Henceforth, we consider five data generating processes with varying degrees of asymmetry and non-convexity in the multivariate response distributions:

1. Multivariate Gaussian:

$$y_n \sim \mathcal{N}(\mu, \Sigma) \text{ with } \mu = [.2, .2, .2]' \text{ and } \Sigma_{jj} = .4, \Sigma_{jk} = .25 (\forall j \neq k).$$

2. Multivariate Student-t:

$$y_n \sim t_r(\mu, \Sigma) \text{ with } r = 5, \mu = [.2, .2, .2]' \text{ and } \Sigma_{jj} = .4, \Sigma_{jk} = .25 (\forall j \neq k).$$

3. Multivariate log-Gaussian:

$$y_n \sim \log \mathcal{N}(\mu, \Sigma) \text{ with } \mu = [.2, .2, .2]' \text{ and } \Sigma_{jj} = .4, \Sigma_{jk} = .25 (\forall j \neq k).$$

4. Conditional heteroskedasticity:

$$y_n \sim \mathcal{N}(\mu, \Omega_n) \text{ with } \mu = [.2, .2, .2]' \text{ and } \Omega_n = \exp(z_n)\Sigma \text{ where } z_n \sim \mathcal{N}(0, 1) \text{ and } \Sigma_{jj} = .4, \Sigma_{jk} = .25, \forall j \neq k.$$

5. Multivariate Gaussian mixture:

$$y_n \sim \mathcal{N}(\mu_{z_n}, \Sigma_{z_n}) \text{ with } \Pr[z_n = m] = \frac{1}{3}, \text{ for } m = 1, 2, 3, \mu_1 = [2, 2, 2], \mu_2 = [0, 0, 0], \mu_3 = [-2, .5, 1] \text{ and } \Sigma_{1,jj} = .4, \Sigma_{1,jk} = .25, \Sigma_2 = I, \Sigma_{3,jj} = .7, \Sigma_{3,jk} = .5.$$

For each data generating process, we generate 1000 data sets with a sample size of 10,000.

We analyze each simulated data set with four methods. (1) The conditional univariate quantiles retrieved from fitting our multivariate Gaussian mixture model to the data. We assume an overfitted mixture distribution of order $M = 5$ and set the hyperparameters for the hierarchical Dirichlet prior on the mixing weights to $\underline{a}_1 = 10$ and $\underline{a}_2 = 40$ to keep the conditional allocation probabilities to spurious components small. The hyperparameters for the Gamma prior on the component prior variance are set to $\underline{b}_1 = .5$ and $\underline{b}_2 = .5$. We keep every 10th of 50,000 MCMC draws after an initial burn-in phase of 10,000. (2) A standard univariate quantile regression model with the linear regression quantiles estimated independently for each response variable. Here, the regression quantiles essentially correspond to the constant. (3) A univariate linear quantile regression model with the other response variables of the output-vector in the conditioning set. (4) A univariate non-linear quantile regression model with the level and squared response variables in the conditioning set.

All models are evaluated on the basis of the average squared deviation of the estimated regression quantiles from the true conditional quantiles for $\alpha \in \{.2, .4, .6, .8\}$. The respective population quantile functions are provided as a supplementary appendix. We use posterior means as point estimates for the Bayesian models. The corresponding simulated mean squared errors are summarized in Table 2. In general, the expansion of the model input-space with the other variables in the output-vector, increases the performance. The multivariate Gaussian mixture model recovers the conditional quantiles of the symmetric uni-/ and the multi-modal settings well

Table 2
Mean squared errors for five simulated multivariate response distributions.

		Gaussian	Student-t	log-Gaussian	conditional heteroskedast.	Gaussian mixture
$\alpha = .2$	QReg	.216	.367	.214	.353	1.960
	QReg with $y_{(-k)}$	.000	.013	.004	.024	.539
	QReg with $y_{(-k)}, (y_{(-k)})^2$	.000	.008	.002	.012	.269
	MQReg	.000	.004	.001	.003	.004
$\alpha = .4$	QReg	.196	.332	.328	.315	1.517
	QReg with $y_{(-k)}$	.000	.001	.007	.002	.487
	QReg with $y_{(-k)}, (y_{(-k)})^2$	.000	.001	.004	.001	.253
	MQReg	.000	.003	.026	.002	.003
$\alpha = .6$	QReg	.197	.330	.492	.315	1.244
	QReg with $y_{(-k)}$	.000	.001	.011	.002	.404
	QReg with $y_{(-k)}, (y_{(-k)})^2$	.000	.001	.005	.002	.247
	MQReg	.000	.003	.048	.001	.003
$\alpha = .8$	QReg	.216	.361	.850	.350	1.084
	QReg with $y_{(-k)}$	.000	.013	.002	.024	.197
	QReg with $y_{(-k)}, (y_{(-k)})^2$	.000	.009	.001	.012	.124
	MQReg	.000	.004	.081	.002	.002

^a QReg: univariate linear quantile regression; QReg with $y_{(-k)}$: univariate linear quantile regression with the other response variables in the conditioning set; QReg with $y_{(-k)}, (y_{(-k)})^2$: univariate non-linear quantile regression with the other level and squared response variables in the conditioning set; MQReg: multivariate Gaussian mixture quantile regression.

^b The reported mean squared errors are averaged across response variables and simulation draws.

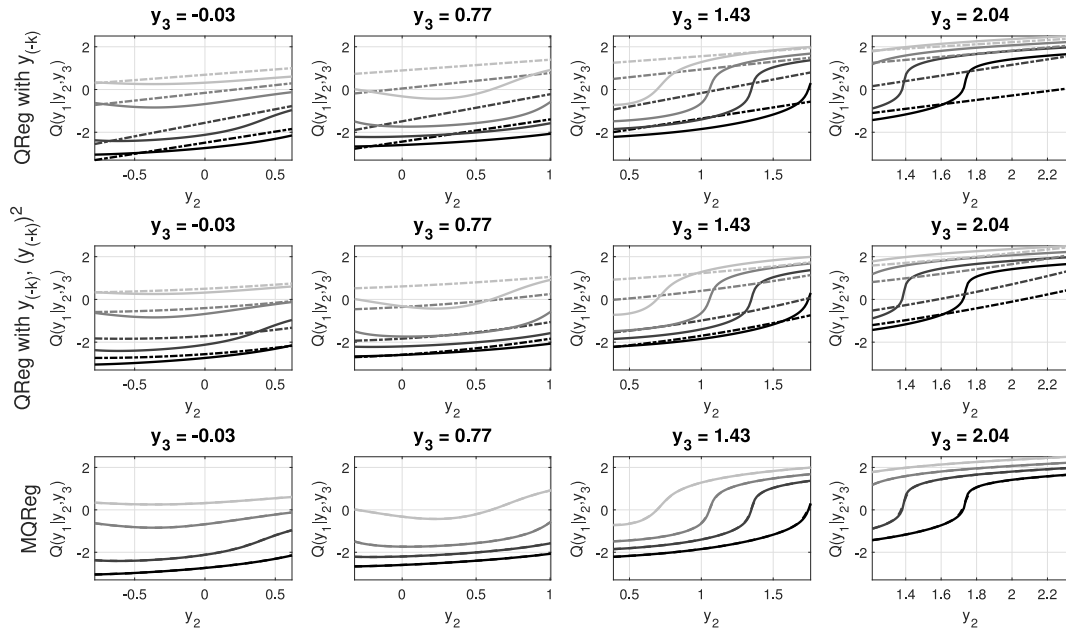


Fig. 3. Estimated (dashed lines) vs. true (solid lines) conditional quantiles of DGP 5, a multivariate Gaussian mixture response distribution, for $\alpha \in \{.2, .4, .6, .8\}$.

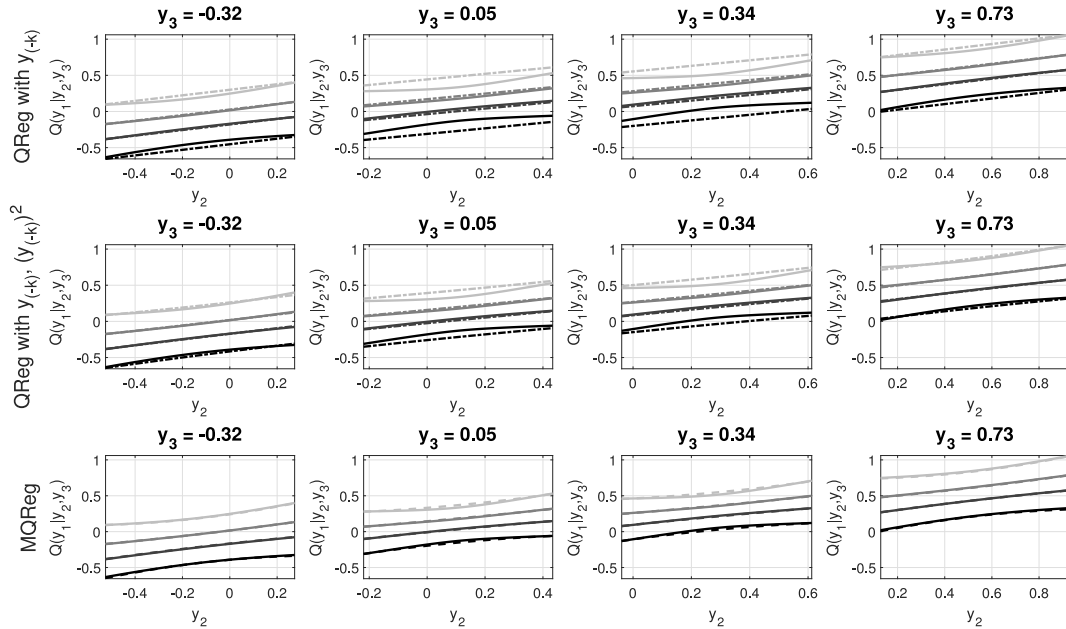


Fig. 4. Estimated (dashed lines) vs. true (solid lines) conditional quantiles of DGP 4, a conditional heteroskedasticity response distribution, for $\alpha \in \{.2, .4, .6, .8\}$.

as compared to the univariate linear and non-linear specifications. Likewise, the asymmetries in the conditional heteroskedasticity design are accurately recovered, particularly in the lowest and highest quantiles of the conditional response distributions. Only for the extremely skewed multivariate log-Gaussian distribution our Gaussian mixture model's fit decreases in the tails.

For DGP 5, the Multivariate Gaussian mixture, Fig. 3 compares the estimated quantiles of response variable y_1 against the true α -level quantiles for different values of y_2 (on the horizontal axis) and y_3 (across the different subplots). The estimated conditional univariate Gaussian mixture quantiles perfectly match the true conditional quantiles, while the conditional univariate linear and non-linear quantiles only give global approximations. Likewise, as illustrated in Fig. 4, the asymmetries in DGP 4, the conditional heteroskedasticity process, are perfectly retained. For the simulated uni-modal and symmetric distributions all three models provide similar estimates (the figures of the remaining simulation designs are collected in a supplementary appendix).

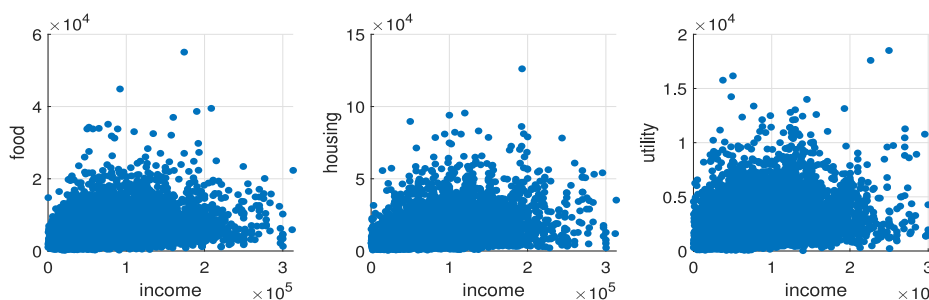


Fig. 5. Empirical distribution of the three expenditures categories food, housing, and utilities conditional on income before taxes.

6. Empirical application

The study of heterogeneity in household consumption patterns is a classical application of quantile regression in economics. In the following, we analyze the variation in annual real expenditures on food, housing and utilities (i.e., fuels, gas, electricity, telephone and public services) jointly with household income. The data are a subset of households from the U.S. Consumer Expenditure Survey collected by the Bureau of Labor Statistics in 2015.² We only consider households consisting of one or two earners, living in an owned or rented property, and having two or less children. This leaves us with $N = 7207$ observations for $K = 4$ response variables. The household income is the amount the household earned in the chosen year before taxes. Fig. 5 illustrates the empirical distribution of the three expenditure categories conditional on income. Note that the sum of the expenditures does not amount to the overall consumption of the household and thus, the considered categories are not complete.

Again, we assume an overfitted mixture distribution of order $M = 5$. We a-priori favor allocations to several components, to address the asymmetries in the empirical distribution, and set the hyperparameter for the hierarchical Dirichlet prior on the mixing weights to $\underline{a}_1 = 10$ and $\underline{a}_2 = 40$. The data is additionally log-transformed in order to fit the income as well as expenditure distributions with fewer mixture components. The hyperparameters for the Gamma prior on the component prior variance are set to $\underline{b}_1 = .5$ and $\underline{b}_2 = .5$. We keep every 40th of 200,000 MCMC draws after an initial burn-in phase of 400,000. All estimates are presented on the original scales. For the multivariate (conditional) superlevel set, we obtain the distribution for the original data from the distribution function of the log-transformed data. For the univariate (conditional) quantiles, we apply the inverse transformation directly to the estimated quantile.

Fig. 6 shows the bivariate SLS contours for expenditures on food and housing ($\mathcal{K} = \{\text{food}, \text{housing}\}$) conditional on the marginal α -level quantiles of income ($\mathcal{C} = \{\text{income}\}$).³ We implicitly marginalize over the utilities expenditures. The blue lines correspond to the bivariate SLS contours at $\alpha \in \{.2, .4, .6, .8\}$, the red lines give the SLS contours conditional on an income increase of \$6000. Conditional on the four income levels, those households with the least food and housing expenditures (.2-SLS, smallest enclosed area) do not react considerably to the income increase for all four income levels. In contrast, those in the highest expenditures for housing and food (.8-SLS, largest enclosed area) do increase their spending substantially after a raise in income. This effect is strongest for the lowest income SLS. Moreover, the multivariate superlevel sets do not indicate clear substitution effects between the two expenditure categories.

Next, we take a univariate perspective. Fig. 7 depicts the quantile-varying marginal effect for a raise in income of \$6000. The associated consumption changes greatly vary along the income distribution: Low-income households dedicate most of the additional income to necessities that relate to spending on food and shelter. In contrast, households in the middle of the income distribution enjoy greater discretionary power and thus, seem to adjust their food and utilities expenditures the most. Households in the highest quantiles of the income distribution hardly increase their spending on food and utilities at all.

7. Conclusion

We present a comprehensive and computationally attractive Bayesian framework that provides a convenient alternative to multiple-output quantile regression models in practice. To that end, we define our object of interest as a property of an (estimated) conditional multivariate density. The multivariate superlevel set (SLS) ensures control over the conditional probability content, the quintessential property for a probabilistic interpretation. It is unique, affine equivariant, and nested across α -levels. The shapes of the SLS contours are flexible and thus, this definition works well even for complex distributions. The Bayesian semiparametric

² The corresponding survey interview files are public available via https://www.bls.gov/cex/pumd_data.htm. The analysis is based on the variable labels: houspq (housing), foodpq (food), utilpq (utilities) and fincbtxm (income before taxes). The replication codes are available at <https://github.com/aschnuecker/Superlevel-sets>.

³ On an Intel(R) Core(TM) i7-8750H CPU 2.20 GHz with 16.0 GB RAM, it takes 4.3 s to calculate conditional quantiles for three variables evaluated at 100 values of the conditioning variables, and for four α -level values. On the same machine, it takes about 35 s for the entire creation of Fig. 6. The estimation of the mixture model takes 153.78 min for 600,000 MCMC draws.

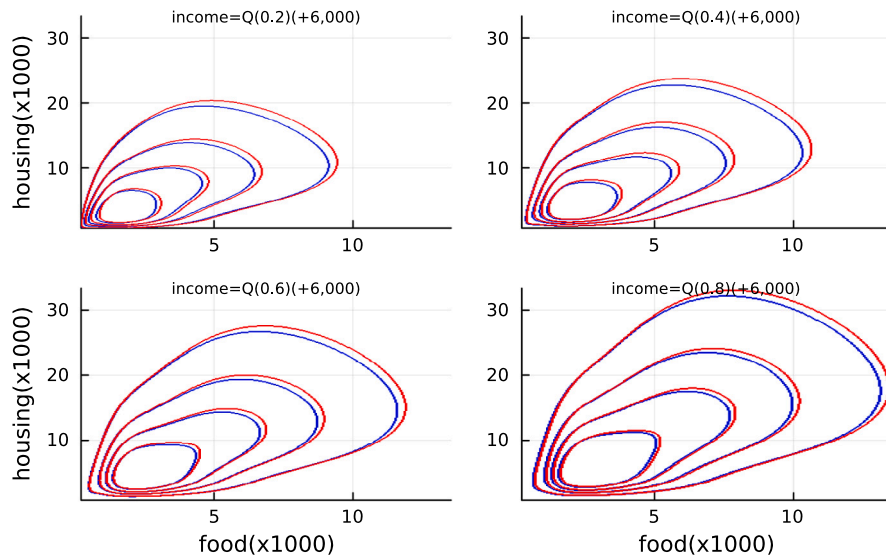


Fig. 6. Bivariate superlevel set contours for the two expenditure categories food and housing conditional on different levels of income. Blue lines (—) correspond to $\alpha \in \{.2, .4, .6, .8\}$ conditional on the indicated income levels, red lines (—) correspond to an income increase of \$6000. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

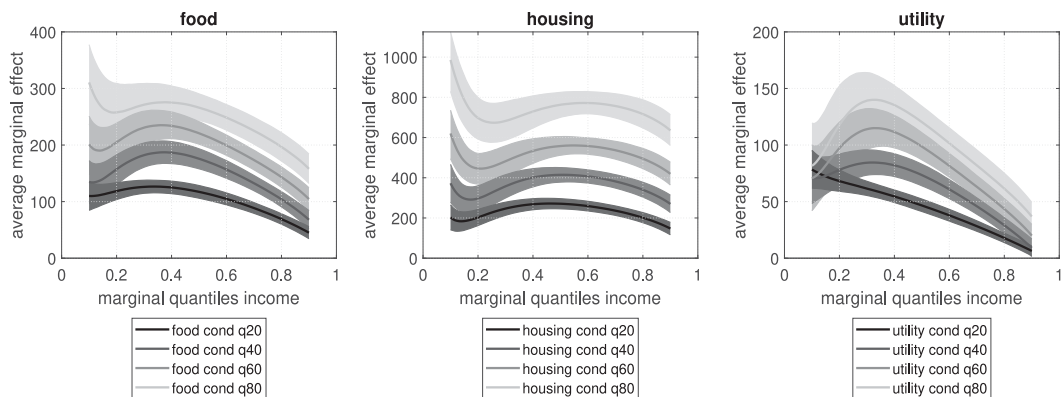


Fig. 7. Quantile-varying marginal effects for the three expenditures categories food, housing, and utilities conditional on different levels of income. Shaded areas indicate the corresponding 90% confidence intervals for four α -level values of income.

density estimation algorithm allows for straightforward incorporation of prior information regarding shapes and centers. We are able to capture linear and non-linear dependencies in the SLS estimates and in the context of univariate quantile regression. Our idea on superlevel sets can be combined with alternative, also non-Bayesian, density estimation approaches.

We obtain the superlevel sets and quantiles directly from the component-specific parameter solutions of a finite overfitted Gaussian mixture model. Since all results are derived from a global posterior, logically consistent (i.e. non-crossing) SLS as well as quantile contours are automatically ensured. Moreover, unpleasant computational features such as runtime issues due to slow convergence rates are avoided (see, e.g., Polonik, 1997). As such, the presented strategies can be applied even to very large and/or multidimensional data sets.

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Appendix A. Equivalence proof

In the following, we prove the equivalence of the superlevel set, and the solution to the problem of finding the smallest coverage probability region, for a random variable with continuous density and unbounded support. Recall the smallest (possible) region definition:

$$Q_Y(\alpha) = \underset{Q}{\operatorname{argmin}} |Q|, \text{ subject to } \Pr[Y \in Q] \geq \alpha.$$

We drop the explicit conditioning to reduce notational clutter. This problem implies that there are no points outside $Q_Y(\alpha)$ with a density larger than $t^* = \inf_{y \in Q_Y(\alpha)} f_Y(y)$. Consequently, t^* characterizes $Q_Y(\alpha)$ and the solution to Eq. (6) coincides with the (strict) superlevel-set of the density at level t^* .

Proof. Denote Q a minimum volume solution to Eq. (6). We show that for a point $y_{\text{out}} \notin Q$ with $t_{\text{out}} \equiv f_Y(y_{\text{out}}) > t^*$, we obtain a contradiction with Q being a solution.

Suppose that such an outside point y_{out} does exist. Let $S_{\epsilon_{\text{out}}}(y_{\text{out}})$ be an area with volume $\epsilon_{\text{out}} > 0$ and with properties: (i) $y_{\text{out}} \in S_{\epsilon_{\text{out}}}(y_{\text{out}})$, (ii) $S_{\epsilon_{\text{out}}}(y_{\text{out}}) \cup Q = \emptyset$, and (iii) $f_Y(y) > t^* - \delta_{\text{out}}$ for all $y \in S_{\epsilon_{\text{out}}}(y_{\text{out}})$ for a suitably chosen $\delta_{\text{out}} > 0$. Continuity of the density function ensures such an area with volume $\epsilon_{\text{out}} > 0$ must exist: at y_{out} the density equals t_{out} and thus, there must be points in the neighborhood of y_{out} where the density exceeds a value (slightly) below t_{out} . For all $\epsilon \leq \epsilon_{\text{out}}$, we can construct areas such that $S_{\epsilon}(y_{\text{out}}) \subseteq S_{\epsilon_{\text{out}}}(y_{\text{out}})$ and $|S_{\epsilon}(y_{\text{out}})| = \epsilon$ with probability content bounded from below: $\Pr[Y \in S_{\epsilon}(y_{\text{out}})] > (t_{\text{out}} - \delta_{\text{out}})\epsilon$.

Next, choose a value $t_{\text{in}} > t^*$ such that an inside point $y_{\text{in}} \in Q$ with $f_Y(y_{\text{in}}) = t_{\text{in}}$ exists. The infimum of the density over the set Q is t^* . Hence, there must be such a t_{in} . For a suitably chosen $\delta_{\text{in}} > 0$, let $S_{\epsilon_{\text{in}}}(y_{\text{in}})$ be an area around y_{in} with properties: (i) $|S_{\epsilon_{\text{in}}}(y_{\text{in}})| = \epsilon_{\text{in}} > 0$, (ii) $S_{\epsilon_{\text{in}}}(y_{\text{in}}) \subseteq Q$ and (iii) $f_Y(y) < t^* + \delta_{\text{in}}$ for all $y \in S_{\epsilon_{\text{in}}}(y_{\text{in}})$. Again, continuity of the density ensures $\epsilon_{\text{in}} > 0$ exists, and areas $S_{\epsilon}(y_{\text{in}}) \subseteq S_{\epsilon_{\text{in}}}(y_{\text{in}})$ with volume $\epsilon \leq \epsilon_{\text{in}}$ have probability content bounded from above: $\Pr[Y \in S_{\epsilon}(y_{\text{in}})] < (t^* + \delta_{\text{in}})\epsilon$.

Finally, set $\epsilon = \min(\epsilon_{\text{in}}, \epsilon_{\text{out}}) > 0$. For $S_{\epsilon}(y_{\text{in}}) \subseteq S_{\epsilon_{\text{in}}}(y_{\text{in}})$ and $S_{\epsilon}(y_{\text{out}}) \subseteq S_{\epsilon_{\text{out}}}(y_{\text{out}})$, consider the open area $Q' = S_{\epsilon}(y_{\text{out}}) \cup Q \setminus S_{\epsilon}(y_{\text{in}})$ with volume equal to Q , by construction. The probability content exceeds $\alpha + \epsilon(t_{\text{out}} - \delta_{\text{out}}) - \epsilon(t^* + \delta_{\text{in}}) = \alpha + \epsilon(t_{\text{out}} - t^* - \delta_{\text{out}} - \delta_{\text{in}})$. This probability exceeds α if $t_{\text{out}} - t^* > \delta_{\text{out}} + \delta_{\text{in}}$. For example, $\delta_{\text{out}} = \delta_{\text{in}} = \frac{1}{3}(t_{\text{out}} - t^*) > 0$ gives Q' with volume equal to Q but with a larger probability content. This implies, we can reduce the volume of Q' without violating the constraint on the probability coverage and obtain a better solution to Eq. (6) than Q . This contradicts the assumption that Q itself is optimal and completes the proof. $\square$

Appendix B. Posterior simulation algorithm

1. Simulate mixture parameters conditional on $\mathbf{z}_n$ ($n = 1, \dots, N$, $m = 1, \dots, M$):

- Sample $\{\kappa_m\}$ from $D(\bar{\rho}_1, \dots, \bar{\rho}_M)$ where $\bar{\rho}_m = \rho_m + N_m$ and $N_m = \#\{n : z_n = m\}$, the number of observations allocated to the m th component.
- Sample $\{\mu_m\}$ from $\mathcal{N}(\bar{\mu}_m, \bar{V}_m)$:

$$\begin{aligned} - \bar{\mu}_m &= \bar{V}_m(\bar{V}_0^{-1}\bar{\mu}_0 + \sum_m^{-1} N_m \bar{y}_m) \text{ with } \bar{y}_m = \frac{1}{N_m} \sum_{n: z_n=m} (y_n - \mathbf{B}_m \mathbf{x}_n). \\ - \bar{V}_m &= (\bar{V}_0^{-1} + N_m \Sigma_m^{-1})^{-1} \end{aligned}$$

- Sample $\{C_m\}$ from $\mathcal{N}(c_m, C_m)$:

$$\begin{aligned} - c_m &= C_m(C_0^{-1}c_0 + \sum_{n: z_n=m} (\mathbf{x}_n \otimes \mathbf{I}) \Sigma_m^{-1} (y_n - \mu_m)) \\ - C_m &= (C_0^{-1} + \sum_{n: z_n=m} \mathbf{x}_n \mathbf{x}_n' \otimes \Sigma_m^{-1})^{-1} \end{aligned}$$

- Sample $\{\Sigma_m\}$ from $\mathcal{IW}(S_m, s_m)$:

$$\begin{aligned} - s_m &= s_0 + N_m \\ - S_m &= S_0 + \sum_{n: z_n=m} (y_n - \mu_m - \mathbf{B}_m \mathbf{x}_n)(y_n - \mu_m - \mathbf{B}_m \mathbf{x}_n)' \end{aligned}$$

2. Sample z_n to classify observations conditional on mixture parameters ($n = 1, \dots, N$):

- $\pi_m \equiv \Pr[z_n = m | y_n, \kappa, \mu, \mathbf{B}, \Sigma] \propto \kappa_m \phi(y_n; g_m(\mathbf{x}_n), \Sigma_m)$.
- Sample $\{z_n\}$ from $\mathcal{M}(\pi_1, \dots, \pi_M)$.

3. Sample hyperparameters:

- Sample $\{\bar{\rho}_m\}$ simultaneously via a random walk Metropolis Hastings step with proposal density $\log(\rho_m) \sim \mathcal{N}(\log(\rho_m), s_{\rho_m}^2)$ from

$$p(\bar{\rho}_m | \kappa) \propto p(\bar{\rho}_m) \frac{\Gamma(M \bar{\rho}_m)}{\Gamma(\bar{\rho}_m)^M} \left(\prod_{m=1}^M \kappa_m \right)^{\bar{\rho}_m - 1}$$

- Sample $\{\lambda_k\}$ from a generalized inverted Gamma distribution $\mathcal{GIG}(\underline{b}_1 - M/2, 2\underline{b}_2, \delta_k)$ where $\delta_k = \sum_{m=1}^M (\mu_{m,k} - \bar{v}_{0,k})^2 / R_k^2$ is the distance between the component locations and prior means.
- Sample $\bar{v}_0$ from $\mathcal{N}(\sum_{m=1}^M \mu_k / M, \bar{V}_0 / M)$ with $\bar{V}_0 = \text{diag}(R_1^2 \lambda_1, \dots, R_K^2 \lambda_K)$.

If no observation is allocated to component m during classification in step 2, all component-specific parameters of this empty component are sampled from their priors.

Appendix C. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.jeconom.2024.105807>.

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